



A step towards cost-effective, environment-friendly Li-ion batteries

Researchers at Ningbo University and University of Puerto Rico-Rio Piedras Campus have developed an ultralow-concentration electrolyte that may be suitable for practical appli-



The newly developed LiDFOB electrolyte for lithium-ion batteries presents a cost-effective, highly conductive and stable alternative to traditional lithium salts, enhancing battery performance and sustainability while simplifying production and recycling processes (Credit: <https://scitechdaily.com>)

cation in lithium-ion batteries. They have published in *Angewandte Chemie* about cells utilising this electrolyte alongside conventional electrodes, showing promising performance metrics. Traditional lithium-ion batteries power various devices but face challenges

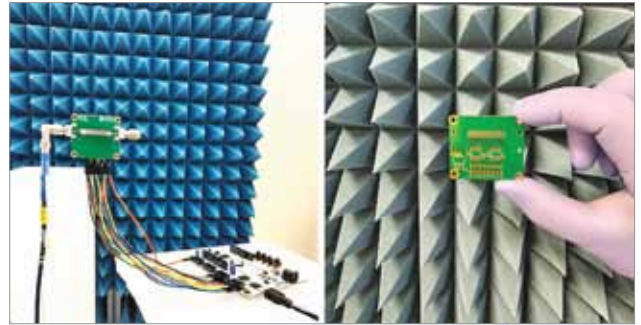
with existing electrolytes like lithium hexafluorophosphate (LiPF₆) due to high viscosity and cost. The LiDFOB electrolyte, with just 2% lithium salt content, offers reduced costs and enhanced stability during production. Its resilience against moisture and air ensures safer manufacturing and recycling processes, aligning with sustainability goals. Leveraging ethyl carbonate/dimethyl carbonate as solvents, the LiDFOB/EC-DMC formulation balances low salt concentration with sufficient ionic conductivity, fostering robust electrode interphase layers for improved cycling stability. This research marks a significant step towards cost-effective and environment-friendly lithium-ion batteries.

Hydrogen storage capacity quadrupled

Hydrogen emerges as pivotal for future energy needs due to its renewable production methods and diverse applications in electricity and heat generation. Industries like steelmaking, glass, and chemicals could benefit, while it promises to revolutionise international shipping and transportation, stabilising power grids by storing surplus renewable energy. Researchers from Skoltech and international partners introduced compounds storing up to four times more hydrogen than current materials, vital for a sustainable, low-carbon economy. Hydrogen storage faces challenges due to its lightness and reactivity, addressed by chemical storage methods. Their study unveils cesium heptahydride (CsH₇) and rubidium nonahydride (RbH₉), surpassing prior materials in storage capacity. Future refinements aim to enhance practicality and scalability, potentially transforming energy systems towards sustainability.

World's first 60GHz mmWave antenna for ISM applications

A research team at the University of Glasgow has developed a revolutionary dynamic metasurface antenna (DMA), the world's first to operate at the 60GHz mmWave



An innovative wireless communications antenna (Credit: <https://www.gla.ac.uk>)

band, reserved for ISM applications. Enabled by specially designed metamaterials, DMA features fully tuneable elements for sophisticated leaky-wave antennas, offering high-frequency reconfigurable operation. The compact prototype incorporates high-speed interconnects for simultaneous parallel control of metamaterial elements, enabling nanosecond beam switching and stable network coverage. DMA holds transformative potential in various fields, with applications in patient monitoring, high-resolution radar, and holographic imaging.

A new power converter with up to 98.3% efficiency

Researchers from Kobe University and National Chung Hsing University have achieved a DC-DC power conversion technology, with potential implications for renewable energy and electric vehicles. Published in the *IEEE Transactions on Power Electronics*, their study unveils a new power converter boasting an impressive efficiency of up to 98.3%, significantly surpassing existing systems. This converter is adaptable, accommodating various energy sources and voltage requirements, crucial for diverse applications like battery systems. It operates bidirectionally, managing energy flow to and from storage solutions like batteries. Noteworthy features include superior voltage ratio, self-balancing inductor currents, and asymmetrical duty limit control, enhancing stability and performance, particularly in electric vehicle-connected DC microgrids. Thorough experiments and analyses validate its design and scalability, highlighting its potential impact on future DC-DC conversion technology for electric vehicles and renewable energy integration.